Lab Report

Operating Systems Laboratory
Assignment 1

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1 Introduction

Basic image processing application in C++ that sharpens a given image in the PPM format.

2 Experiments

The program was executed five times for each image, and the reported timings represent the average execution time across the five runs.

2.1 Data and Plot

Image	Resolution	Read (μs)	S1 (μs)	S2 (μs)	S3 (μs)	Write (μs)
1.ppm	450×180	7535.0	49634.0	4781.0	3540.6	3338.0
2.ppm	786×393	19527.8	159127.2	12362.4	7795.2	7173.2
3.ppm	1024×575	26247.8	236684.2	23307.0	15265.8	13741.6
4.ppm	1280×910	57834.2	479908.8	45203.8	29521.2	28068.2
5.ppm	1500×1000	96534.6	605348.2	59899.8	39985.2	37036.4
6.ppm	2271×1500	188778.0	1361542.0	116052.8	86742.4	96533.4
7.ppm	3750×2470	421592.8	3668677.2	313588.2	232831.2	236164.0

Table 1: Image processing timings for PPM images. All times are in microseconds.

S1 = Smoothing, S2 = Details, S3 = Sharpening.

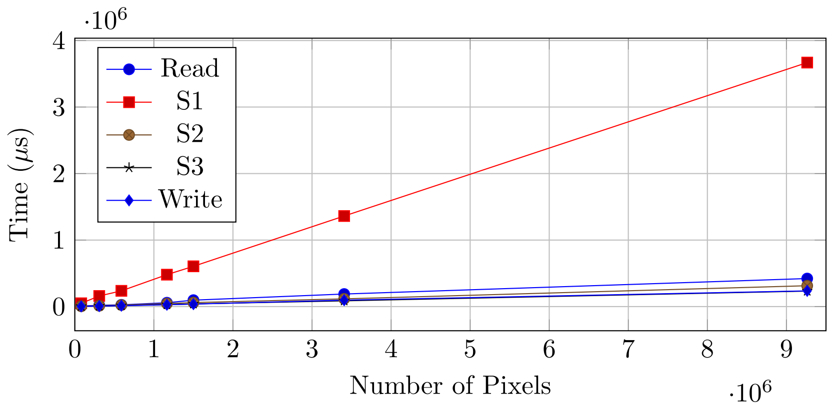

Figure 1: Processing time as a function of image size.

2.2 Inference

Smoothing(S1) is the most computationally intensive stage of the pipeline and its execution time increases approximately linearly with the number of pixels. The remaining processing stages, including I/O, have significantly lower execution times and remain below one second even for the largest input image.